

Literature Status for the Canonical Inclusion Question in arXiv:1811.01265

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Status

This note records that Question 6.2 of Albiac–Ansorena–Cuth–Doucha, arXiv:1811.01265, is answered positively by the later paper Albiac–Ansorena, arXiv:2603.28193.

No new theorem is claimed here. The point of this packet is to clear an open-problem signal from the run queue and identify the exact later theorem which settles it.

Source Question

In Section 6 of arXiv:1811.01265, after proving that canonical maps need not be isometric, the authors ask whether the weaker isomorphic-embedding conclusion always holds:

Let $0 < p < 1$ and $\mathcal{N} \subseteq \mathcal{M}$ be two p -metric (or metric) spaces in inclusion. Is the canonical linear map of $\mathcal{F}_p(\mathcal{N})$ into $\mathcal{F}_p(\mathcal{M})$ an isomorphic embedding?

This is labelled Question 6.2 in the published version cited by the later paper.

Later Answer

The introduction of arXiv:2603.28193 explicitly cites this as AACD2020, Question 6.2, states that it had remained open beyond the metric case, and says that the paper gives a positive solution in full generality.

The theorem-level statement is arXiv:2603.28193, Theorem 3.13:

for every $0 < p \leq 1$ there is $C(p) < \infty$

such that, whenever $(\mathcal{M}, d)$ is a pointed p -metric space and $0 \in \mathcal{N} \subset \mathcal{M}$, the canonical linearization

$$L[\mathcal{N}, \mathcal{M}; p]: \mathcal{F}_p(\mathcal{N}) \longrightarrow \mathcal{F}_p(\mathcal{M})$$

is an isomorphic embedding and

$$\|L[\mathcal{N}, \mathcal{M}; p]^{-1}\| \leq C(p).$$

This directly answers the source question in the affirmative. The source question also mentions ordinary metric spaces; these are included under the source convention for p -metric spaces when $0 < p \leq 1$, since if d is a metric then d^p is again a metric.

Scope

This packet only records the literature answer to the canonical inclusion question. It does not address the neighboring classification question in arXiv:1811.01265 asking which p -metric spaces force the canonical map to be isometric for every subspace.

References

1. F. Albiac, J. L. Ansorena, M. Cuth, and M. Doucha, *Lipschitz free p -spaces for $0 < p < 1$* , arXiv:1811.01265.
2. F. Albiac and J. L. Ansorena, *Lipschitz extensions into p -Banach spaces, and canonical embeddings of Lipschitz-free p -spaces for $0 < p < 1$* , arXiv:2603.28193.